

Data Center Sustainability: Revisits and Outlooks

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Abstract—As energy-intensive entities, data centers are associated with significant environmental impacts, making their sustainability a subject of growing interest in recent years. In this article, we revisit data center sustainability and propose a forward-looking vision for improving data center sustainability. We argue that data center sustainability encompasses more than just energy efficiency and must be evaluated and optimized through a multi-faceted approach. To this end, we first present an overview of the sustainability metrics from five aspects. After that, we demonstrate the sustainability status of the latest data centers utilizing publicly available data center sustainability ratings. Furthermore, we examine the evolution of data center sustainability standards in Singapore to highlight several trending features. Based on the analysis, we identify several key elements of sustainable data centers. We then propose the Cognitive Digital Twin (CDT) architecture, which incorporates a digital twin engine for system-wide simulation and a decision engine for optimal control to improve data center sustainability. A case study is performed to optimize the chiller plant efficiency of a production data center in Singapore. The results demonstrate that the CDT can improve chiller plant energy efficiency by 5%, indicating around 140 metric tons of annual carbon emission savings.

Index Terms—Sustainability metrics, data centers, green computing, digital twins.

I. INTRODUCTION

TO DATE, with the unprecedented growth of the data center industry, the sustainability issue of data centers has also attracted widespread attention, ranging from the industry to

the government. As the key supporting infrastructure of the fourth industrial revolution and the digital economy, data centers enable the deployment and development of cutting-edge technologies such as 5 G, Artificial Intelligence (AI), Metaverse and etc. Meanwhile, the sustainability of data centers becomes a concerning issue due to their massive energy consumption and the corresponding huge amount of carbon emissions. It is estimated that data centers contribute to 2-3% global electricity consumption, and 0.4-0.75% global carbon emissions [1]. To mitigate the negative environmental impacts of data centers and improve their overall sustainability, several cloud service providers such as Google, Microsoft, and Amazon have released their ambitious goals toward carbon-neutral cloud service. From the government side, data center operators from European Union (EU) have recently committed to the Europe Green Deal to regulate themselves to achieve the EU climate-neutral goal by 2030 [2].

In order to optimize the sustainability of a data center, the first and foremost step is to leverage the properly defined sustainability metrics or the sustainability rating framework to evaluate its sustainability status and identify the direction where the optimization should be conducted. In the past decade, numerous metrics have been proposed to evaluate the performance of a data center from various aspects. In this regard, Reddy et al. have contributed a comprehensive survey on the metrics for data centers from various aspects, including energy, cooling, green, etc [3]. However, despite these contributions, a clear set of representative values for these metrics remains elusive. Besides standalone metrics, there are several data center sustainability rating systems to evaluate data center sustainability. In this regard, the Leadership in Energy and Environmental Design (LEED) is the most prevailing green building certification framework in North America, aiming to comprehensively evaluate the sustainability of a building from various aspects including energy efficiency, site sustainability, water efficiency and etc [4]. It is designed for rating all kinds of buildings besides data centers. A data center will earn a certain amount of scores by achieving some design or operational requirement. Another example is the Green Mark rating system developed by Singapore Building and Construction Authority. It is a dedicated sustainability rating system for data centers, and it evaluates a data center from five aspects including energy efficiency, cooling efficiency, airflow management and etc [5]. Despite the availability of these ratings, a comprehensive analysis of the sustainability status of the data center industry and evolving trends has yet to be conducted.

In this study, we aim to assess the current state of data center sustainability and introduce a transformative solution for

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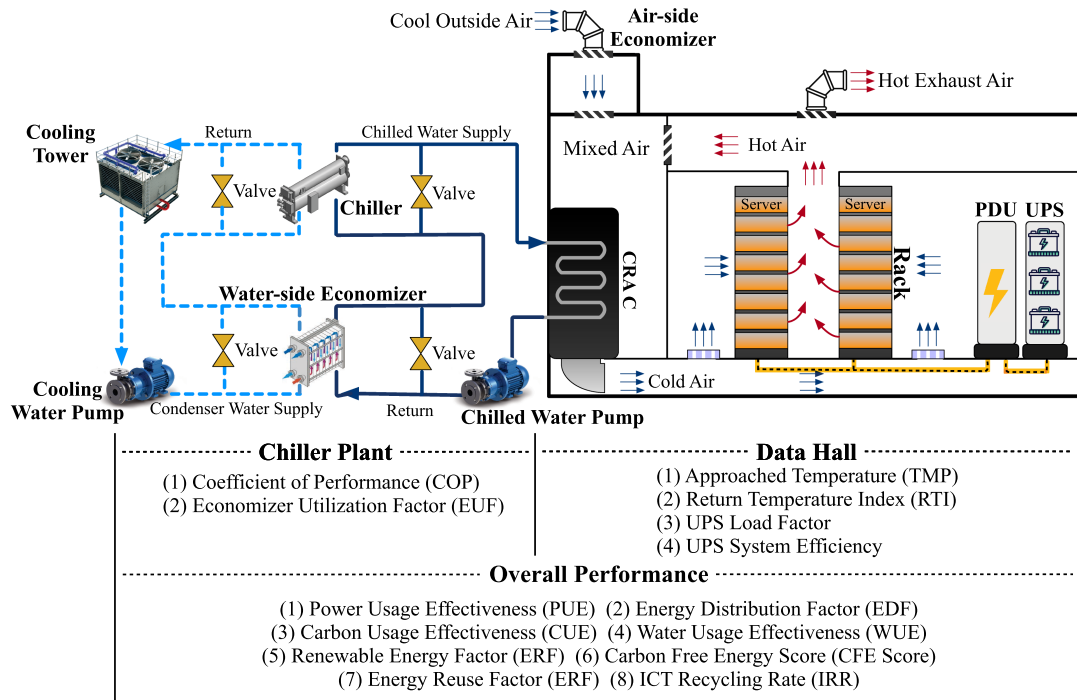


Fig. 1. A typical data center containing the IT equipment (servers, storage devices, and networking devices) and the facility to deliver necessary cooling and electrical power. Critical sustainability metrics are curated correspondingly.

promoting data center sustainability through the Cognitive Digital Twin (CDT) approach. To do this, we first provide an overview of the critical metrics for evaluating the sustainability of data centers. Then, we analyze the sustainability ratings of the latest data centers to determine their sustainability status. Moreover, we examine the progression of data center sustainability standards in Singapore to shed light on the emerging trends in optimizing data center sustainability. Based on our analysis, we highlight the key components of sustainable data centers in the future. Finally, we propose the architecture of the CDT to support the rapid transformation toward sustainability in the data center industry. To demonstrate the effectiveness of the CDT, we apply it to a production data center in Singapore to optimize the performance of its chiller plant. The results show that the CDT is capable of not only predicting the energy consumption of the chiller plant accurately but also reducing carbon emissions by approximately 140 metric tons per year.

Paper Organization. In Section II, we introduce the metrics for evaluating data center sustainability. In Section III, we conduct analysis on the data center sustainability rating results as well as the evolution of data center sustainability standards in Singapore to reveal some trending features for sustainable data centers. Based on the analysis, we identify several key design elements for sustainable data centers in Section IV. Subsequently, we introduce the architecture of our CDT platform in Section V and conduct a case study to validate its practical applicability. Finally, we conclude by providing several envisions on sustainable data centers.

II. REVISIT OF DATA CENTER SUSTAINABILITY METRICS

In this section, the focus is on the essential metrics for ensuring the sustainability of data centers. A typical data center is illustrated in Fig. 1 and consists of IT equipment and the supporting infrastructure that provides energy, cooling, and other resources. The electrical power is sourced from the power grid and flows into the Power Distribution Unit (PDU) and the Uninterrupted Power Supply (UPS), which distributes the power to the IT equipment. Cooling for the IT equipment is provided by the Computer Room Air Conditioning (CRAC) unit and the chiller plant. The IT equipment, including servers, networking devices, and storage devices, is housed in server racks that are arranged in rows and are responsible for providing uninterrupted cloud services. The sustainability metrics can be categorized into five classes: energy efficiency, airflow management, cooling efficiency, electrical power supply system efficiency, and environmental impacts (See Table 1).

A. Energy Efficiency Metrics

Since data centers are energy-intensive enterprises, it is pivotal to improve their energy efficiency to avoid energy waste and boost sustainability. Energy efficiency is usually evaluated by the useful work done by a component or the whole data center divided by the total energy delivered to it. In the past decades, a plethora of energy efficiency-rated metrics was proposed for data centers to evaluate energy efficiency in the different hierarchies, ranging from server level to facility level [3]. Among them, the most recognized metric is the Power Usage Effective (PUE)

TABLE I
EXISTING OPERATIONAL SUSTAINABILITY METRICS FOR GREEN DATA CENTERS

Metrics Category	Key Metrics	Unit	Typical Value (Range)
Energy Efficiency	• PUE • EDF	• Ratio • Percentage	• 1.55 [1] • NA
Airflow Management	• TMP • RTI • RCI	• °C • Ratio • Ratio	• NA • 80%-120% [7] • 90%-100% [8]
Cooling Efficiency	• COP • Economizer Utilization Factor	• Ratio; • Ratio	• 2.0-7.0 [9] • NA
Electrical Power Chain Efficiency	• UPS Load Factor; • UPS System Efficiency	• Ratio; • Ratio	• NA • 0.86-0.95 [10]
Environmental Impacts	• CUE • WUE • REF • CFE Score • ERF • IRR	• kgCO₂/kWh; • Litre/kWh; • Ratio; • Percentage; • Ratio; • Ratio	• NA • 1.8 [11] • NA • NA • NA • NA

These metrics can be categorized into 5 classes, i.e., energy efficiency, thermal and airflow management, cooling efficiency, electrical power chain efficiency, and environmental impacts.

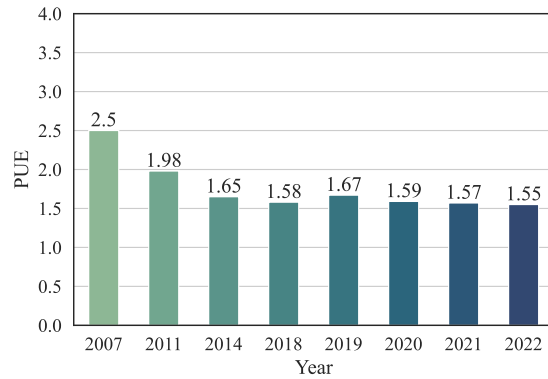


Fig. 2. Average PUE of global data centers from 2007 to 2022. Substantial improvement in terms of PUE can be observed in the last decade. Further optimization of the PUE is stagnant.

proposed by Green Grid [6], and it is widely adopted in the sustainability rating systems like the Green Mark [5] and the LEED [4]. PUE is defined as the ratio of the annual total energy consumption and the annual IT equipment energy consumption. Obviously, the range of PUE is between 1.0 and infinity, and a lower PUE indicates that a higher ratio of energy is delivered to the IT equipment to produce useful work and less energy is used to cool down the IT equipment or deliver electrical power to the IT equipment. Since the birth of the PUE, substantial efforts in the data center industry have been made to improve it and Fig. 2 shows its trending in the past decade [1]. It can be seen clearly that the average PUE has been optimized significantly and reached a plateau of 1.55 in 2022. Currently, the continuous optimization of PUE seems to be stagnant and it also implies that it is time for the data center industry to think beyond energy efficiency to achieve further sustainability improvement. Furthermore, it is also worth noting that the PUE metric is biased and the hyper-scale data centers will gain unfair advantages over the small-scale data center since the IT energy consumption in the hyper-scale data centers is significantly larger, probably leading to a smaller PUE value. Therefore, the PUE is not suitable for comparing the energy efficiency between data centers with significant capacity differences.

Another important metric for evaluating data center energy efficiency is the Energy Distribution Factor (EDF) which reflects the distribution of energy consumption of different non-IT components (the cooling system, the UPS power loss, the lightning system and etc). With the EDF metric, data center operators can identify the inefficient components in the data center, and perform optimization over these components to improve overall energy efficiency in a more pertinent way.

B. Airflow Management Metrics

Air management is an essential aspect of sustainability improvement and risk mitigation in data centers. Poor air delivery can result in local hot spots, causing unexpected server shutdowns and increased energy consumption, as server energy consumption rises with the ambient temperature. This section will explore metrics to evaluate the efficiency of air delivery and thermal management in data centers.

The effectiveness of airflow delivery in a data center is indicated by the Approached Temperature (TMP), which is the difference between the supply temperature from the CRAC unit and the return temperature. A small TMP indicates opportunities for optimizing airflow and improving energy efficiency. From Fig. 3, we can see that a larger TMP indicates that the CRAC fan speed can be reduced, resulting in energy savings in the cooling process.

Another two commonly used metric in evaluating airflow management efficiency is the Return Temperature Index (RTI) and the Rack Cooling Index (RCI) [8]. RTI measures the relationship between the Approached Temperature and the difference between the average server outlet temperature and the server inlet temperature. Ideally, RTI is 100% when the CRAC return temperature matches the average server outlet temperature and the CRAC supply temperature equals the average server inlet temperature. A value of less than 100% for RTI suggests that some cold air is not being captured by the CRAC unit's return, resulting in a lower return temperature. On the other hand, a value greater than 100% means that some hot air from the server outlet is being recycled back to the server inlet, causing the inlet temperature to rise. Using the RTI metric, data center operators can improve the separation of hot and cold airflows, thereby

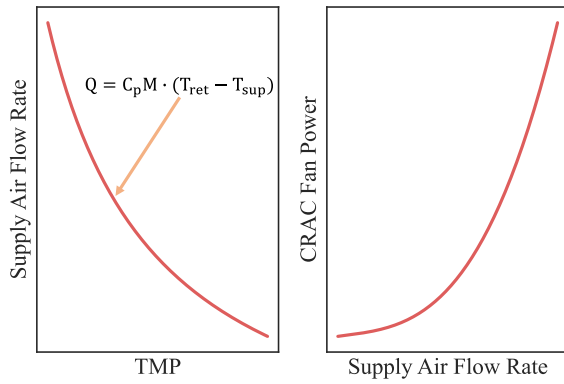


Fig. 3. Illustration of the relationship between the approached temperature (TMP) and the supply air flow rate when the heat load in the data hall is fixed. With a higher TMP, the supply air flow rate can be reduced, thus saving the cooling energy.

allowing for an increase in supply air temperature, which reduces the power consumption of the cooling system. As for the RCI, it is commonly used to evaluate compliance with the ASHRAE data center thermal guidelines [8]. RCI is 100% if all rack intake temperatures fall within the recommended temperature range recommended by ASHRAE. However, if some rack intake temperatures exceed the maximum recommended temperature, the RCI value decreases. To maintain a safe and efficient data center, it is recommended to keep the RCI value above 96% [8].

C. Cooling Efficiency Metrics

The cooling system plays a critical role in the data center operation and its efficiency is also strongly related to the overall sustainability of a data center because the cooling system accounts for 30-50% energy consumption of a data center [12]. Several metrics have been proposed for assessing the cooling system efficiency in the past decade. Among them, the most commonly used metric is the Coefficient of Efficiency (COP), which is the ratio of the useful cooling energy delivered by the cooling system and the network put into it. It is suitable for both the water-cooled chilled water cooling system and the air-cooled chilled water cooling system, and the energy consumption of all components in the data center cooling system such as the pump, the chiller, etc should be taken into account. Higher COP indicates that less energy is consumed by the cooling system to provide the required cooling load for the data center, indicating that the cooling system is more efficient and sustainable. Currently, the typical value range of COP is between 2.0 and 7.0 [9].

Besides COP, with the development of free cooling technology, an increasing number of data centers are equipped with air-side or water-side economizers in their cooling system to provide free cooling, and several metrics for evaluating the efficiency of such cooling systems are emerging recently. The most popular metric for this purpose is the Economizer Utilization Factor (EUF), which is the total hours when the economizer is operating divided by the total hours in the accounting period. Obviously, the unit of this metric is a percentage and it has an ideal value of 100% which implies that all the cooling load is fulfilled with

the economizer. By tracking this metric, a data center operator can develop a suitable mechanism to take full advantage of the free cooling capacity offered by the local weather and improve overall sustainability.

D. Electrical Power Supply System Efficiency Metrics

The efficiency of the electrical power supply system is also crucial for data center sustainability. This system ensures uninterrupted operation, but when it is oversized, there can be a significant amount of energy loss due to no-load usage. The UPS Load Factor is a metric used to determine if the UPS system is oversized by measuring the peak load in comparison to its design capacity. A low UPS Load Factor indicates that the system may be oversized and result in energy loss.

To further evaluate the energy loss in the UPS system, one can use the UPS System Efficiency metric, which is calculated by dividing the output power by the input power. The typical value for this metric ranges from 0.86 to 0.95 [10]. The higher the value, the more efficient the UPS system is and the less energy is lost. Monitoring the UPS System Efficiency can provide valuable insight into the status of the UPS system, allowing the data center operator to conduct predictive maintenance and improve sustainability.

E. Environmental Impacts Metrics

Massive energy usage and carbon emissions of data centers have raised global concern about their environmental impacts and a couple of environmental impact metrics were proposed. These metrics can be grossly categorized by the following concepts: *reduce, renewable, recycle*.

The environmental impacts of a data center mainly come from the carbon footprint due to massive energy drawn from the power grid and the water usage in the chiller plant to provide the necessary cooling. In this regard, two metrics are proposed by Green Grid to guide the data center to *reduce* its carbon footprint and water usage. The first metric is the Carbon Usage Effectiveness (CUE) [13], which is the ratio of the total carbon emissions and the IT equipment energy consumption. When all energy comes from the electrical power grid, the CUE is equal to the carbon emission factor of the local grid multiplied by the PUE of the data center. When part of the total energy comes from renewable sources, the corresponding carbon emissions should be subtracted from total carbon emissions. Similar to the CUE, Water Usage Effectiveness (WUE) [14] is used for evaluating the effectiveness of a data center in its water usage. It is defined as the ratio of the total site water consumption and the total IT electrical energy consumption and the typical value of this metric in the United States is 1.8 according to a report from Lawrence Berkeley Lab in 2016 [11].

Another way to reduce the overall environmental impacts is to take advantage of *renewable* energy. In order to monitor the usage of renewable energy and maximize its utilization, several metrics have been proposed. One metric is called the Renewable Energy Factor (REF), which describes the percentage of renewable energy in the total energy consumption of a data center. REF is an effective metric to monitor renewable energy usage and

encourage the diversification of the data center energy portfolio by increasing the usage of renewable energy. However, it only accounts for the total renewable energy percentage in a time period instead of real-time tracking. To further reveal the renewable energy utilization status, the data center operator can choose to use the Carbon Free Energy Score (CFE Score), which evaluates the degree to which the energy consumption of a data center is matched by the CFE on an *hourly* basis [15]. By using this metric, data center operators can discover the energy mismatch in real-time and schedule the available resource subsequently (IT task, cooling capacity, etc) so that renewable energy utilization can be maximized.

Besides *reduce* and *renewable*, it is also worth considering *recycling* to minimize the overall environmental impacts of a data center. One critical metric to evaluate energy reuse efficiency is the Energy Reuse Factor (ERF), which is the ratio of total energy reused and the total energy consumption of a data center. Higher ERF indicates that more waste energy is harvested for alternative purposes. Another important metric is the IT Recycling Rate (IRR) which quantifies to what extent the decommissioned IT equipment is disposed of in a “responsible” way. “Responsible disposal” means that the decommissioned IT equipment is either reused or recycled in a responsible way. By improving IRR, the overall environmental impacts of a data center can be further reduced because more IT equipment is recycled instead of being sent to landfill.

III. REVISIT OF DATA CENTER SUSTAINABILITY STATUS AND TRENDS

In this section, we conduct two studies to reveal the sustainability status and trends in the data center industry. We start with the analysis of the sustainability rating scores of the data center projects in the LEED database. Then, we analyze the evolution of Singapore’s data center sustainability standard.

A. Data Center Sustainability Rating Analysis

We herein study sustainability scores of the data centers which take the LEED sustainability evaluation framework. The LEED rating score is public available.¹ We only choose the data center projects which take the latest LEED BD+C protocol to expose the sustainability status.

To start with, we analyze the categorical score distribution of the LEED rating result. Because the energy efficiency score accounts for the largest percentage, we herein illustrate the energy efficiency score and other scores separately, and the result is illustrated in Fig. 4. It can be seen from Fig. 4 that the data centers which are granted as Gold or Platinum have better energy efficiency scores than the Silver ones in general. Therefore, improving energy efficiency is the most straightforward approach to boosting the overall sustainability of a data center. However, it can also be observed that some data centers with desirable energy efficiency scores still fail to be certificated as Gold or Platinum. For those data centers, we can find that they perform poorly

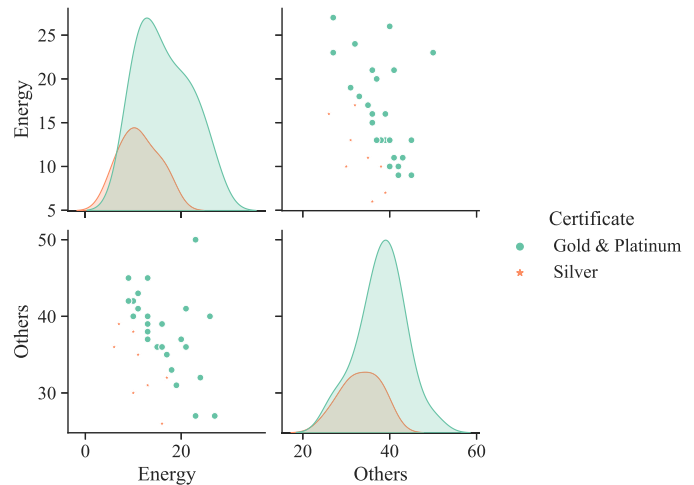


Fig. 4. Illustration of the distribution of energy efficiency scores and sum scores of other evaluating categories for data center projects in the LEED database.

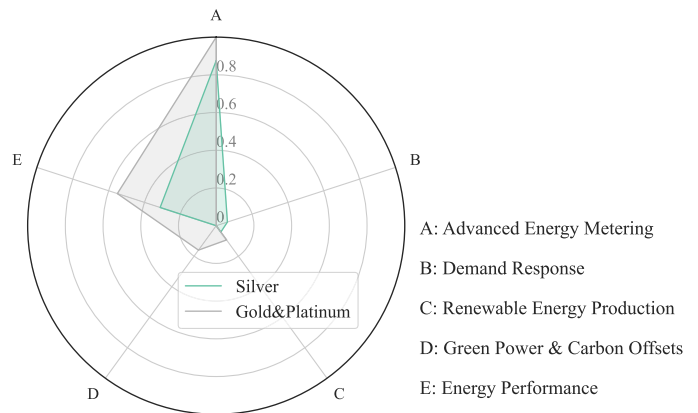


Fig. 5. Illustration of the average score for each credit element in the energy efficiency category for data center evaluated with the LEED. Most data centers have installed advanced energy metering systems.

in other aspects (e.g., water usage efficiency). Therefore, data center sustainability is beyond standalone energy efficiency and a multi-pronged approach should be taken to optimize overall sustainability.

We further investigate the average score of each credit item in the energy efficiency subcategory to understand current bottlenecks in improving data center energy efficiency. Fig. 5 shows that most data centers perform very well in terms of energy metering. This observation is inspiring because advanced energy metering infrastructure can provide data center operators with valuable insights into the deficiency in the energy supply system so that they can take proper actions to boost energy efficiency. However, we can also observe that most data centers do not have satisfactory energy performance. There might exist various reasons for this observation including inefficient IT equipment, improper provisioning of IT equipment, lack of IT-facility co-optimization, etc. To improve energy efficiency, joint IT-facility optimization via state-of-the-art deep reinforcement learning is

¹ Rating results available in <https://www.usgbc.org/projects?Search+Library=%22data+center%22>, access date: 2022-8-14.

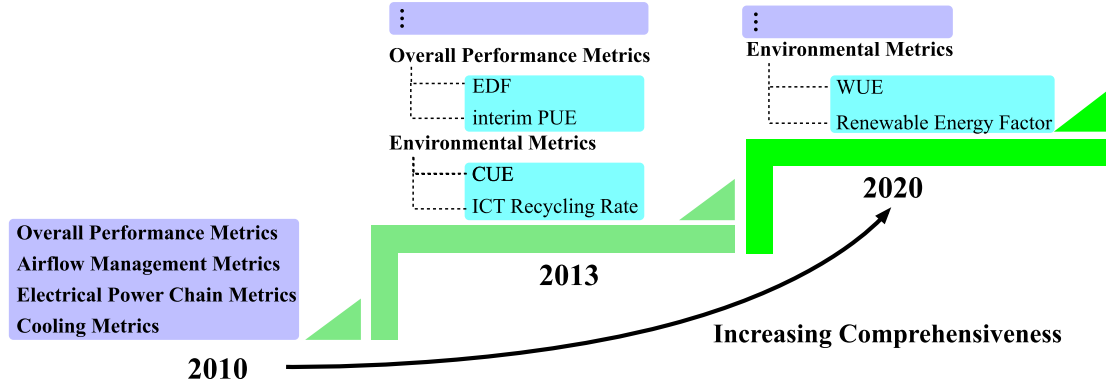


Fig. 6. Evolution of Singapore Sustainability Standard for green data centers (SS564). In the past decade, the environmental impacts of data centers are receiving increasing attention.

a promising approach [16]. Furthermore, Fig. 5 also reveals the risk-averse mindset in the current data center industry in utilizing renewable energy and carbon offsets to lower carbon footprint as most data centers receive zero marks in renewable energy production and green energy utilization. It is understandable since data centers are mission-critical infrastructure and renewable energy may be extremely intermittent, which impedes the uninterruptible operation of data centers. In this regard, some pioneering cloud providers such as Google have attempted to maximize the utilization of green energy by carbon-aware computing server provision [17]. However, pushing such a transformation in the data center industry is difficult because the management of a green energy-integrated data center is in its infancy and more efforts should be devoted to accelerating the green transformation.

B. Evolution of Singapore's Sustainability Standard

In this section, we present the evolution of the Singapore Standard for green data centers² to expose the trends in boosting data center sustainability. The evolution is illustrated in Fig. 6.

The first trend is that the latest sustainability standard calls for *higher granularity* in information revealing. It can be seen from the comparison between the 2013 version and the 2020 version that the latter augments the overall performance metrics with the interim PUE which requires PUE reporting for a shorter period of time. Therefore, new data centers should install an advanced energy metering system to report and store energy consumption data at higher frequencies, even in real-time. Therefore, high information revealing granularity will be the prerequisite for future sustainable data centers. Furthermore, comparing the 2013 version and the 2010 version, it can be observed that the former adds the EDF metric to encourage the reveal of the energy consumption of each subsystem in the data center. This regulation puts forward higher requirements for device-level energy usage information reveal, highlighting the necessity of the advanced energy metering system.

²<https://www.imda.gov.sg/regulations-and-licensing-listing/ict-standards-and-quality-of-service/IT-Standards-and-Frameworks/Green-Data-Centre-Standard> access date: 2022-8-14

The second trend is the increasing focus on data center *environmental impacts*. Environment impact indicators including the CUE and the IRR are first included in the standard are first included in 2013, indicating the special attention to carbon emissions incurred by massive energy consumption, as well as the environmental impacts from landfilling the decommissioned IT equipment containing a huge amount of heavy metals. Furthermore, another two environmental impact indicators, i.e., the WUE and the REF, are taken into account in 2020, showing increasing emphasis on overall sustainability improvement. By adding the REF metric, existing data centers with renewable energy-compatible electrical systems will be encouraged to utilize on-site or off-site renewable energy to enrich their energy supply portfolio and reduce their environmental impacts. Moreover, new data centers will be incentivized to install renewable energy-compatible electrical systems to improve both their design and operation sustainability. Similarly, the newly-added WUE metric will also motivate data center designers to implement water recycling technologies in order to improve water reuse and reduce potable water consumption, leading to a more sustainable and environmental-friendly data center.

IV. OUTLOOK OF KEY DESIGN ELEMENTS FOR NEXT-GENERATION SUSTAINABLE DATA CENTERS

Based on the sustainability status and trends presented previously, we discuss several key design elements of a sustainable data center in this section, including energy-efficient IT equipment and service, advanced physical and cyber infrastructure, renewable energy integration, carbon measurement and reporting, as well as carbon-neutral cloud services.

A. Energy-Efficient IT Equipment & Cloud Services

IT equipment creates the demand for electrical power and cooling in the data center, any energy efficiency improvement of IT equipment will have a huge impact on the overall sustainability of a data center. To improve data center energy efficiency, one can seek hardware solutions and software solutions.

On the hardware side, data center operators are encouraged to install energy-efficient IT equipment in the data center. For

example, as suggested in [18], data center operators should procure IT equipment with ENERGY STAR certification. This equipment has installed an advanced power management system that supports dynamic power capping, leading to 30% energy savings compared with standard servers.³ Furthermore, a sustainable data center can also adopt heterogeneous servers to improve energy efficiency. In addition, innovative server-level direct liquid cooling can offer additional energy savings by removing the dissipated heat from chips using the liquid coolant with higher heat exchange efficiency. Direct liquid cooling might offer significant advantages in very high energy-intensive scenarios such as high-performance computing. Moreover, it also offers the opportunity for harvesting the server waste heat for additional energy-saving [12].

On the software side, data center operators are encouraged to use software control technologies such as virtualization, server consolidation, and task dispatch to reduce the energy footprint of the IT equipment by provisioning computing resources in an on-demand manner. In this regard, many previous works studied how to judiciously assign computing tasks to servers by considering energy consumption [19], [20], [21]. The rationale of such solutions is that current data centers host heterogeneous computing devices with diverse energy efficiency and computing power. Meanwhile, data center workloads are also different in terms of resource demand and quality-of-service (QoS) requirements. Therefore, by allocating different workloads to different computing devices properly, significant energy efficiency gains are expected. Besides computing task scheduling, at the individual server level, power mode management and server CPU frequency scaling can also be incorporated to further reduce server energy consumption [22], [23], [24]. Moreover, if we further consider the data center network, up to 28.37% energy saving is available if the Software Defined Network (SDN) is deployed to effectively reduce the usage of active hosts in a cloud data center [25].

B. Advanced Infrastructure

Advanced infrastructure, including the physical and cyber one, serves as a key enabler for boosting the sustainability of a data center. In the physical domain, data center operators should focus on optimizing the physical configuration such as installing the hot/cold aisle containment and the return plenum, designing raised floor, device segregation, etc. For example, a data center with the hot/cold aisle containment will be rewarded in the Green Mark rating system [5] and it is also recommended in [18]. With the containment, the chance of hot air re-circulation will be greatly reduced, making it possible for raising the supply air temperature to reduce cooling energy consumption.

As for the cyber domain, energy and water usage metering systems and Data Center Infrastructure Management (DCIM) system are credited to facilitate resource usage tracking and optimization. For water metering, the basic requirement is to report the water consumption at the building level *monthly* or *annually* to the corresponding authorities [4]. To support water

management and identify opportunities for additional water savings, data center designers are also encouraged to install water meters at the cooling towers to meter replacement water use of all cooling towers [4]. Similar to water metering, the basic requirement for energy metering is to report the total energy consumption at the building level. To achieve further energy saving, data center designers should consider installing advanced sensor network-based energy metering systems for all components in a data center including servers/racks, CRACs, electrical systems, etc. Furthermore, it is required that the metering system should be able to report energy data at any frequency and store historical data for at least 36 months [4]. Building upon the metering system, the DCIM system is a necessary ingredient for improving data center sustainability. It is used for displaying, tracking, monitoring, and optimizing the energy consumption as well as efficiency of each component in a data center. In the Green Mark rating system [5], data centers are credited for installing DCIM systems that can monitor energy consumption, and IT equipment utilization, and perform power control such as power capping or software control such as virtualization to reduce energy consumption.

C. Renewable Energy Integration

Recently, renewable energy is becoming an indispensable element for sustainable data centers as more and more cloud providers start to integrate renewable energy into the energy supply to their data centers [26]. Therefore, major data center sustainability rating systems have taken renewable energy production and utilization into consideration to boost the utilization of renewable energy as an alternative energy option for data centers. To smoothly transform to renewable energy-powered data centers, several challenges should be addressed in next-generation sustainable data centers.

First, renewable energy, e.g., solar energy and wind energy, is intermittent, calling for advanced energy supply-demand coordination. In this regard, many previous works focused on IT workload scheduling and planning to improve renewable energy utilization [27], [28]. Although IT workload management can better leverage renewable energy, it will inevitably cause performance degradation and job delay.

Second, intermittent renewable energy incurs significant difficulties in providing a resilient power supply, resulting in a higher risk of a power outage. According to the latest survey from Uptime Institute [1], power is still the main cause of the outage, and the cost of each outage can exceed \$1 million. To reduce the risk, energy storage devices, such as batteries and UPS, should be considered for sustainable data center operations. With energy storage, surplus renewable energy can be stored for future usage, which offers more flexibility to IT workload management and can effectively reduce performance degradation [29], [30], [31]. However, the charging and discharging of energy storage devices will shorten their lifespan. It will lead to a higher total cost of ownership (TCO) as well as carbon footprints as the energy storage devices are still expensive and their embodied carbon footprint will be accounted into a data center's SCOPE 3

³https://www.energystar.gov/products/enterprise_servers

emissions.⁴ Thus, judiciously using energy storage devices by jointly considering monetary cost, component lifespan, availability of renewable energy, and characteristic of IT workloads should be carefully considered in future sustainable data centers.

D. Carbon Measurement & Reporting

As the environmental impacts of data centers have been a major concern in sustainable development, measuring and reporting them will be mandatory for future sustainable data centers. However, as stated in the latest survey from Uptime Institute, only 37% of the surveyed data center operators report their carbon emissions [1]. This will soon become an area of concern as most data centers will be required to provide such data under the new regulations that are being implemented around the world. For the carbon emission reporting, data center operators are encouraged to account for their Scope 1 and 2 emissions⁵ at the beginning and extend the reporting to Scope 3 emissions in the future.

To achieve accurate carbon measurement, the data center operators should understand the source of carbon emissions and how they are accounted for. Carbon emissions from data centers can be categorized into embodied carbon emissions and operational carbon emissions. The embodied emissions refer to those from the manufacturing of servers and physical facilities. The operational emissions, which contribute the majority of carbon emissions, come from the electricity consumed by data centers. To measure the total carbon emissions from a data center, operators should take the local grid carbon intensity into account to report the CUE or carbon emissions by multiplying carbon intensity with the total energy consumption. Furthermore, cloud providers are also encouraged to reveal the carbon emissions from running a computing instance, especially for the computational-intensive AI model training tasks. In this regard, various carbon footprint calculators for public cloud services are available, e.g., [32].

E. Carbon Neutral Cloud Services

As large-scale cloud data centers represent huge loads on the electrical grid and the decarbonization of cloud services should be accelerated to mitigate their environmental impacts. As the carbon intensity and the presence of renewable energy in regional electrical grids are different, the cloud service provider can design elaborate task schedulers to maximize the utilization of renewable energy or minimize the carbon emissions of a network of data centers. As an exemplary implementation, Google just proposed the Carbon Intelligent Computing Platform (CICS) for a large-scale data center which schedules computing capacities *temporally* by considering the future carbon intensity of the local grid [17]. In this thread, other researchers considered a data center with renewable energy facilities and schedule the delay-tolerant computing tasks temporally according to the presence of renewable energy to mitigate

the environmental impacts [26]. Furthermore, some researchers explore the computing task scheduling problem in the context of *geographically distributed* data centers to minimize the carbon emissions of cloud services [33], [34], [35]. We refer interested readers to [12] for a more detailed discussion about the design principle of such a carbon-aware scheduler.

V. COGNITIVE DIGITAL TWINS FOR DATA CENTER SUSTAINABILITY OPTIMIZATION

With the rapid expansion of large-scale data centers, traditional model-based or rule-based data center operation policies are not sufficient for sustainable transformation in the data center industry. On the other hand, learning-based operation policies have become a strong competitor in data center operations due to their capability of handling sophisticated dynamics within a large-scale data center [12]. In the latest survey from Uptime Institute, 57% of the surveyed data center operators will trust a fully-trained AI model in data center operations [1]. To fully train an AI model, we need to collect sufficient data covering various situations, which is prohibitive because data centers are mission-critical facilities and their safe operation is first priority. To bridge this gap, we propose the Cognitive Digital Twin (CDT), a system that encapsulates the digital twin engine that can simulate the full system states and the data-driven decision engine to derive optimal control policy. We first articulate the architecture of the CDT and then validate its practical applicability via a case study on the chiller plant COP optimization of a production data center in Singapore.

A. Architecture of the Cognitive Digital Twin

The architecture of the proposed CDT is shown in Fig. 7, consisting of the digital twin engine, the decision engine, and the application layer. In the application layer, we would support various functions including PUE/CUE optimization, sustainability reporting, what-if analysis as well as green financing. The two supporting pillars are the digital engine and the decision engine, which will be introduced in detail in the rest of this section.

The digital twin engine is one of the technological cores of the system and it serves as a virtual testbed with which the learning-based controller interacts. To be specific, we will design a simulation engine to collect the different simulators from different disciplines together to perform a sophisticated multi-physics simulation. With this high-fidelity interconnected digital twin platform, we can provide fine-grained simulation functionality for each individual component of a data center. For example, with the coupled data hall twin and the heating, ventilation, and air conditioning (HVAC) twin, we can investigate how the temperature field within a data hall evolves with specific setpoints provided by the HVAC system as well as the system energy consumption under the control policy. By doing so, the learning agent can derive optimal policy by interacting with the coupled digital twins while guaranteeing no local hot spots will occur.

The digital twin engine relies on three key steps: modeling, calibration, and simulation. The modeling procedure automatically generates a new simulation model (e.g., a computational

⁴Scope 3 emission comes from the entire value chain of the data center.

⁵Scope 1 emission refers to the emissions from direct consumption of on-site fossil fuel combustion for generating electricity. Scope 2 emissions come from the grid electricity usage.

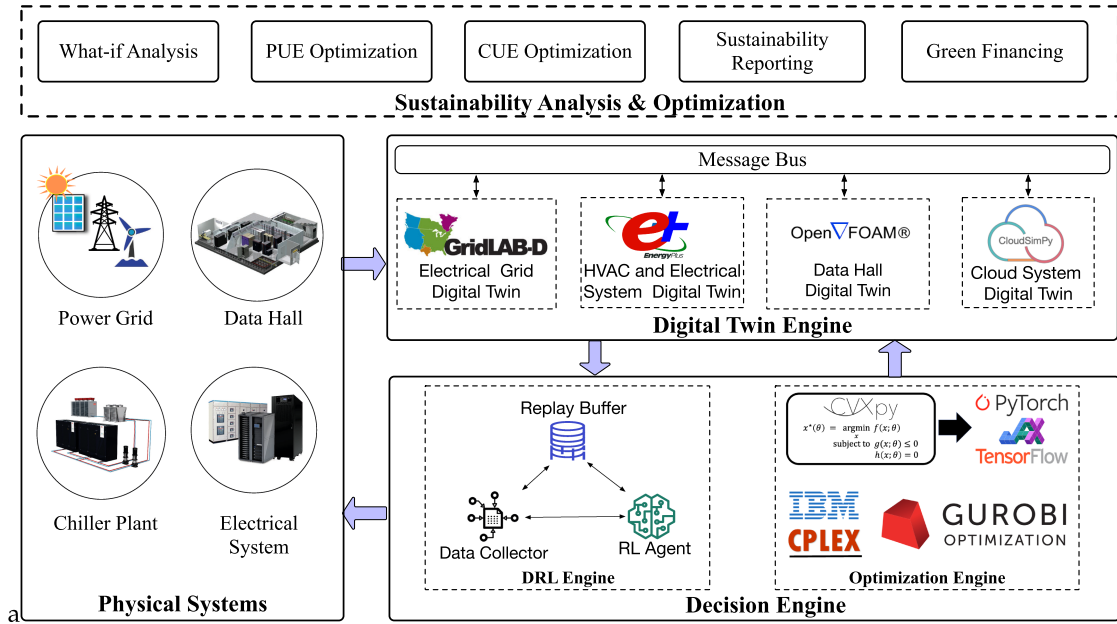


Fig. 7. Architecture of the Cognitive Digital Twin (CDT). The core modules are the digital twin engine and the decision engine. The digital twin engine encapsulates various simulators from different disciplines to conduct a system-wide simulation to obtain the full system state. The decision engine consists of a Deep Reinforcement Learning (DRL) engine and a numerical optimization engine to derive optimal control policies. With the cognitive digital twin, data center operators can conduct holistic sustainability analysis to understand the bottleneck in sustainability improvement and conduct optimization.

fluid dynamics (CFD) model for the data hall) by parsing the design data via a dedicated parser. However, the inevitable modeling error and inherent non-stationary nature of the physical data center necessitates the user updating the model regularly to bridge the gap between digital and physical systems. The digital twin engine conducts auto-calibration to improve the accuracy based on operational data, eliminating the need for time-consuming and labor-intensive manual operations [36]. The engine will then perform system-wide simulations to obtain the system states (e.g., temperature, energy consumption, electricity price) under hypothetical boundary conditions and use them for failure analysis, predictive maintenance, and control optimization. The system-wide simulation will face two major challenges: 1) computation overhead mismatch; 2) design of interface data. For the first challenge, the data hall based on the CFD simulation uses the numerical algorithm to solve for the temperature/velocity/pressure of each spatial grid for a target data hall with millions of grids. Therefore, the CFD simulation requires intensive computation. In contrast, as the heat extraction process and power distribution process do not involve solving a complicated partial difference equation, their solving processes only incur light computation. To bridge this gap, alternative surrogate models to accelerate the CFD simulation should be considered. In this regard, a highly accurate physics-guided surrogate model for the CFD simulation has been proposed and integrated into the digital twin engine to realize the real-time simulation of the temperature distribution within a data hall. For the second challenge, we propose adopting the *external coupling* scheme to achieve multi-physics simulation in a minimal intrusion manner [37]. Specifically, external coupling exchanges system state data (e.g., HVAC supply setpoints, return

temperature, etc) according to the physical properties of the coupled domains periodically. In this way, different simulation programs remain independent and no further modification of the source code of these simulators is required.

The decision engine is the other technological core in the CDT system, aiming to search for the optimal control policy to maximize data center sustainability by interacting with the digital twin engine. The core module in the AI engine is a Deep Reinforcement Learning (DRL) library, which implements data collectors, trainers, replay buffers and RL policies (e.g., PPO [38], PADQN [39] etc). Moreover, the diverse digital twins' system dynamics make DRL training difficult. For example, the control period for the cooling and IT system may differ by an order of magnitude. As a result, training a DRL agent to coordinate the cool control and IT scheduling is non-trivial. In this regard, a cooperative and interactive Multi-Agent Reinforcement Learning (MARL) approach is envisioned to train these AI agents effectively. In addition, since the policy learning algorithms might involve complicated numerical optimization processes (e.g., Trust Region Policy Optimization (TRPO) [40]), we also implement an optimization engine to support the solving of a typical optimization problem (e.g., convex quadratic programming). Furthermore, the optimization engine can also be the supporting pillar for some safe reinforcement learning algorithms [41], [42], [43] to ensure safe exploration and learning in the online phase.

B. Case Study: Chiller Plant Optimization

The cognitive digital twin has been deployed in a large-scale production data center in Singapore and demonstrated its strong

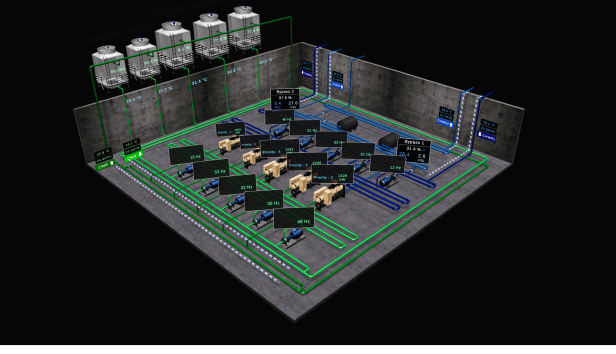


Fig. 8. The 3D model for the chiller plant of the production data center in Singapore. The chiller plant has 5 chillers to provide chilled water to extract heat from the data halls.

capability in optimizing the energy efficiency of the data center chiller plant system. According to the result reported in the latest survey paper [12], the chiller plant system contributes around 70% energy consumption of the data center cooling system. Therefore, the energy efficiency of the chiller plant will have a huge impact on the data center's sustainability. However, current data centers often utilize the rule-based policy to operate the chiller plant. Such a policy heavily relies on the expertise of the data center operator and is not flexible in dealing with dynamic IT workloads. To this end, we leverage our cognitive digital twins to search for the optimal control policy for the chiller plant system while respecting data hall thermal compliance simultaneously.

We use EnergyPlus [45], a well-established building modeling and simulation tool, to build our chiller plant energy model. In our case, the physical chiller plant hosts 5 chillers, 5 cooling towers, 1 chilled water pump, and 1 condensing water pump. In addition, there is a bypass loop in the physical system. The bypass loop is constructed to allow the chiller plant to supply more chilled water than the actual requirement from the data hall to reduce the data hall thermal safety risk. However, any over-supplied chilled water will pass through the bypass loop, resulting in energy efficiency loss. Therefore, in our case, we also want to regulate the bypass flow rate lower than a threshold δ . There are 5 data halls in the data center, hosting thousands of servers with a 5 MW rated IT load. We also build a 3D model to visualize the operation status of each component in the chiller plant as shown in Fig. 8.

To make the energy model a digital twin, we first conduct the model calibration using real sensor data with a 1-minute sampling interval collected from the DCIM system of the data center. The sensor readings include the IT heat load in each data hall, the chilled water supply temperature, and the flow rate, as well as the energy consumption of each component in the chiller plant (chillers, pumps, cooling towers, etc). In the calibration, we use the historical sensor readings of the chiller water temperature, chilled water flow rate, and the IT heat load of each data hall as input to the energy model, which produces the energy consumption of the chillers, the pumps as well as the cooling towers. We manually adjust the energy model parameters to make the predicted energy consumption close to

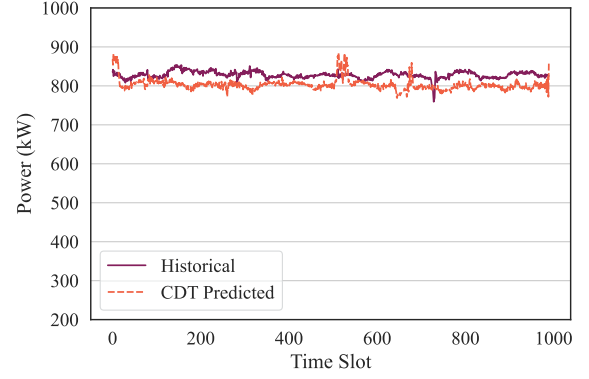


Fig. 9. Chiller plant total energy consumption calibration results for the data center in Singapore. The mean relative error in energy consumption prediction is 3.52%, which is acceptable according to the ASHRAE standard [44].

the historical sensor readings. The calibration result is illustrated in Fig. 9. With the calibration, the mean relative error for the hourly chiller plant energy consumption prediction is 3.52%, which is acceptable according to the ASHRAE standard [44].

With the calibrated digital twin, we train a DRL agent by interacting with it to learn the optimal control policy of the chiller plant. The system states, control actions and the reward function are defined as follows.

- **Actions:** At the k -th control period, we define the actions applied to the actuators as $\mathbf{a}[k] = [T_{\text{chw}}[k], M_{\text{chw}}[k]]$, where $T_{\text{chw}}[k]$ and $M_{\text{chw}}[k]$ denotes the chilled water supply temperature and mass flow rate at the k -th time step.
- **States:** The system states at the k -th control period contains the IT heat load and the room temperature in each data hall at each time step, i.e., $\mathbf{s}[k] = [\mathbf{P}_{\text{IT}}[k], \mathbf{T}_z[k]] \in \mathbb{R}^{10}$. Here, $\mathbf{P}_{\text{IT}}[k] \in \mathbb{R}^5$ and $\mathbf{T}_z[k] \in \mathbb{R}^5$ represent the IT workload power consumption and the data hall air temperature of each data hall at the k -th time step.
- **Reward:** We choose the COP as the reward that we want to maximize. The COP is defined as the ratio between the total cooling load of the chiller plant and the total IT power consumption, i.e., $r[k] = \frac{Q_{\text{cool}}}{\sum_i P_i}$. The total cooling load Q_{cool} can be calculated via our energy model.

In our case, since we jointly control the chilled water supply temperature and flow rate, it is possible that the data hall air temperature cannot be maintained at the desired setpoint due to the insufficient chilled water supply from the chiller plant. We herein define a cost when the zone air temperature is above the desired setpoint $\bar{T}$ (25 °C). The cost function is defined as $c_z(T_z[k]) = \text{clip}(T_z[k]; \bar{T}, \epsilon)$, where $\text{clip}(\cdot)$ denotes the clip function that returns a non-zero value for any value not in the range $[\bar{T} - \epsilon, \bar{T} + \epsilon]$ and the ϵ represents the maximum allowable zone air temperature fluctuation. The cost is proportional to the amount of violation. According to the service level agreement (SLA), we set $\epsilon = 1$ in this case. A cost will also be issued when the bypass flow rate is higher than the threshold, i.e., $c_b(M_{\text{chw}}[k]) = \max\{M_{\text{chw}}[k] - \delta, 0\}$. In this case, we set the threshold for bypass flow rate $\delta = 10$. To explicitly take the constraints into consideration, we formulate the chiller plant optimization problem as a Constrained Markov Decision

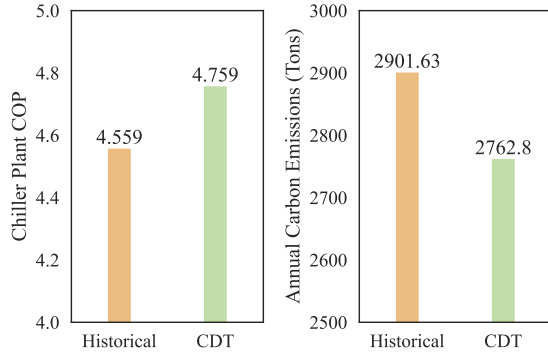


Fig. 10. Chiller plant COP and estimated annual carbon emissions for the data center in Singapore before and after optimization. With CDT, we can improve the chiller plant COP by around 5%, implying around 140 metric tons of annual carbon emission savings.

Process (CMDP). To be specific, we would like to find a policy π_θ that is parameterized by θ , so that the cumulative reward is maximized while all cumulative constraints are also satisfied. In our case, we use a feed-forward neural network with three layers to parameterize the policy. The CMDP problem is formally defined in the following:

$$\underset{\theta}{\text{maximize}} \quad \mathbb{E}_{\mathbf{P}_{\Pi}} \left[\sum_{k=1}^{\infty} \gamma_r^k r[k] | \pi_\theta \right], \quad (1a)$$

$$\text{subject to} \quad \mathbb{E}_{\mathbf{P}_{\Pi}} \left[\sum_{k=1}^{\infty} \gamma_c^k c_z^i[k] | \pi_\theta \right] \leq \bar{c}_z, i = 1, 2, \dots, 5 \quad (1b)$$

$$\mathbb{E}_{\mathbf{P}_{\Pi}} \left[\sum_{k=1}^{\infty} \gamma_c^k c_b[k] | \pi_\theta \right] \leq \bar{c}_b. \quad (1c)$$

Here, $\bar{c}_z$ and $\bar{c}_b$ denote the threshold for the constraint violation cost for the zone air temperature and the bypass flow rate. We set $\bar{c}_z = 1$ and $\bar{c}_b = 0.1$ in our case study. The discount factor for the reward process (γ_r) and cost process (γ_c) are set to 0.99.

We leverage the state-of-the-art DRL algorithm, i.e., PCPO [46], in our AI engine to solve the CMDP problems. We use the historical IT heat load of each data hall to drive the training, and the evaluation result is demonstrated in Fig. 10. With the cognitive digital twin, we can improve the chiller plant COP by around 5%, from 4.559 to 4.759. By considering the grid carbon intensity in Singapore,⁶ it implies around 140 metric tons of annual carbon emissions saving. Through this case study, we demonstrate the workflow of our CDT solution for data center sustainability improvement and reveal its potential in dealing with the real-world complicated data center environment.

VI. CONCLUSIONS AND OUTLOOKS

In this article, we provide a revisit and outlook for sustainable transformation in the data center industry. We first categorize

the sustainability metrics to guide the operators to evaluate sustainability comprehensively. We further conduct quantitative studies to reveal the sustainability status of the data center industry based on publicly available data center sustainability ratings. Moreover, several trending features for boosting data center sustainability are also pointed out to facilitate future optimization. Based on the analysis, we identify several critical elements in sustainable data centers. Furthermore, we propose to accelerate sustainability transformation with the CDT framework and conduct a case study using real-world data collected from a production data center in Singapore. The evaluation results demonstrate that around 140 metric tons of annual carbon emission savings can be achieved with the CDT system. By summarizing the review and analysis, we conclude with several envisions.

- Data center sustainability is beyond energy efficiency. The sustainability of a data center should also take water efficiency, environmental impacts, and circular economy into consideration.
- Rethinking the importance of IT infrastructure efficiency. Despite IT equipment being the biggest energy consumer, the unified metrics for evaluating the efficiency of IT equipment remain undefined in the prevailing data center sustainability rating frameworks.
- Renewable energy integration is important. Future sustainability rating systems should put more credits in renewable energy utilization to encourage green transformation in the data center industry.
- Higher information revealing granularity. Data center operators shall reveal their operating performance in more detail. To support this, a networked sensing infrastructure with an advanced DCIM system is preferable for future sustainable data centers. With such a system as the supporting infrastructure, the CDT platform is envisioned to optimize data center sustainability systematically.

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⁶Singapore's grid carbon intensity is around 0.40 kg/kWh according to <https://www.ema.gov.sg/singapore-energy-statistics/Ch02/index2> access date: 2022-12-27.

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